

Prefatory Clarification on the Variational Section

This model intentionally adopts a quasi-solution $\phi(r)$, rather than an exact solution. The reason is that an exact solution would eliminate the physical significance of the characteristic scale $r = R$, thereby preventing the reproduction of the essential behavior: scale-responsive structure of space.

If an exact solution were enforced, the scale $r = R$ would likely be absorbed into coordinate redefinitions or symmetry-induced degeneracies. In such a case, R would lose its status as a physically anchored scale and instead become a redundant parameter, thereby obscuring any meaningful scale-dependent structure in the resulting dynamics.

To verify the validity of this quasi-solution, we conducted a variational residual analysis. The results show that the violation of the extremum condition is localized, well controlled, and appears as a physically meaningful structure. This behavior is also visually illustrated in Fig. 1.

The quasi-solution and the variational residual do not signify compromise or defect in any sense. Rather, they constitute the very foundation of the structure developed here and serve as the key to realizing scale-dependent spatial dynamics.

What is important is that this quasi-solution is not adopted as an approximation, but is introduced for its structural role in deriving the effective Λ term, $\Lambda_{\text{eff}}(r)$.

It does not imply that the entire structure of the universe or the fundamental field equations themselves are quasi-solutions.

The aim of this study is to examine whether such a quasi-solution structure can function effectively within a universe that is otherwise described by exact solutions.

For clarity, the term “logarithmic-type” used in this paper does not refer to a strict logarithmic function itself, but rather to a qualitative feature indicating that the functional form exhibits a logarithmic behavior.

Japanese version

https://misaki-kasai.github.io/paper/f_Lambda_commentary_ja.pdf

Repulsive Gravity Theory

Based on Scalar Fields and the Effective Λ

Misaki Kasai

May 1, 2026

Abstract

In this study, we propose an extension of the Einstein field equations by introducing a repulsive gravity framework based on the scale-responsive geometric repulsion effect of space, derived from a scalar field. This effect is induced by the geometric configuration of the scalar field, from which a scale-dependent effective cosmological term, Λ_{eff} , emerges. The impact of Λ_{eff} depends on spatial scale, potentially redefining the structural properties of gravity, especially at galactic scales and within black hole (BH) interiors. We demonstrate that two key observational and theoretical challenges—the flattening of galactic rotation curves in their outer regions and the avoidance of black hole singularities—can be addressed in a unified and quantitative manner through the common scalar-field-derived term Λ_{eff} .

1 Overview: Galactic Rotation and BH Singularities

Recent astronomical observations have revealed that the rotational velocities of stars in the outer regions of galaxies significantly exceed those predicted by general relativity, even when the observed amount of baryonic matter is taken into account. This discrepancy is known as the *galactic rotation curve problem* and has long been considered difficult to explain without invoking the existence of dark matter [1, 2].

On the other hand, gravitational collapse inside black holes is known, in the classical framework, to lead to a singularity—i.e., a region of infinite curvature—in the limit $r \rightarrow 0$. This phenomenon is considered to signal the breakdown or limitation of general relativity itself [3]. The so-called *singularity problem* strongly suggests the necessity of quantum corrections or structural modifications within gravitational theory.

2 Successes and Limitations of the Λ CDM Model

The Λ CDM model, which serves as the standard framework in modern cosmology, has achieved remarkable success in explaining observational data such as the cosmic microwave background

(CMB), baryon acoustic oscillations (BAO), and Type Ia supernovae with high precision [4]. However, the cosmological constant Λ is defined as a static and uniform quantity, without incorporating any spatial structure or scale dependence.

Moreover, although Λ CDM relies on invisible components such as dark matter and dark energy, the true nature of these entities remains unknown. They function as auxiliary terms in the theoretical framework, and there has been persistent criticism that they lack a solid basis for physical reality [5].

3 Foundational Standpoint

This study attempts to construct a gravitational theory in which **repulsive response is the subject**, in contrast to general relativity, which treats “gravity as attraction (a manifestation of spacetime curvature)” as its primary perspective. Specifically, we propose a framework that introduces an effective cosmological constant term, $\Lambda_{\text{eff}}(r)$, derived from a scalar field. This enables a unified description of gravitational structures ranging from galactic scales to black hole singularity scales. Moreover, in this study, the geometric repulsive response is constructed on the basis of a static, spherically symmetric scalar-field Lagrangian, with the aim of later extending the model to cosmology by regarding the geometric scale ratio $\frac{r}{R}$ as the basis of time evolution in the background universe.

Thus, in the present study, from the standpoint of adopting the one-dimensional geometrical repulsive response structure as the first principle, four-dimensional spacetime and matter are regarded as secondary entities generated from that structure, while covariance is positioned as a means for describing them locally. Within this framework, the cosmological constant Λ is retained in the Einstein field equations as an approximately constant background scale, thereby preserving the philosophical foundation of the Λ CDM model while overcoming its structural limitations.

4 The Scalar Field and the Effective Λ

In this section, we demonstrate that the geometric effective Λ term

$$\Lambda_{\text{eff}}(r) = \frac{4r^2}{(R^2 + r^2)^2}$$

is structurally derived from the variational structure of a spherically symmetric and static scalar field configuration.

Here, R represents the characteristic length scale of each physical regime, while both R and r are treated as dimensionless quantities until calibrated against observations.

The central principle of this study is that the essential repulsive response structure is encoded in a radial one-dimensional sector with characteristic scale R .

Accordingly, we adopt the following basic radial one-dimensional scalar-field Lagrangian for a spher-

ically symmetric and static configuration:

$$\mathcal{L}_\phi = -\frac{1}{2} \left(\frac{d\phi}{dr} \right)^2 - V(\phi(r)),$$

where $V(\phi(r))$ is the scalar potential.

This radial one-dimensional formulation serves to generate a scale-dependent repulsive response in a spherically symmetric and static configuration.

In this study, the fundamental radial variable is not r itself, but the dimensionless scale ratio

$$x \equiv \frac{r}{R},$$

where R is the characteristic scale defined above. To require a regular response structure without singularities on the real axis, we impose the following conjugate pair in the complex x -plane, symmetric with respect to the real x -axis:

$$\zeta_\pm = x \pm i, \quad x \geq 0.$$

By Euler's formula, these can be expressed in polar form as

$$\zeta_\pm = \sqrt{1+x^2} e^{\pm i\theta}, \quad \theta = \arctan\left(\frac{1}{x}\right) \quad (x > 0), \quad \theta(0) = \frac{\pi}{2}.$$

We then define the background profile as the real amplitude response obtained after phase cancellation:

$$\mathcal{B}(x) \equiv \ln \zeta_+ + \ln \zeta_-.$$

Substituting the conjugate polar forms into this definition, we obtain

$$\mathcal{B}(x) = \left[\ln \sqrt{1+x^2} + i\theta \right] + \left[\ln \sqrt{1+x^2} - i\theta \right] = 2 \ln \sqrt{1+x^2} = \ln(1+x^2).$$

Therefore, the logarithmic background profile used in this theory is obtained as the real amplitude response generated through the phase cancellation of the complex-conjugate scale pair. Accordingly, the physical meaning of $\mathcal{B}(x)$ is that it represents a dimensionless geometrical background response that space exhibits with respect to the scale ratio $x = r/R$.

At the characteristic point $x = 1$, namely $r = R$, we have

$$\zeta_\pm = 1 \pm i,$$

and hence

$$\arg \zeta_\pm = \pm \frac{\pi}{4}.$$

Since the phase difference of the complex-conjugate pair is generally $\Delta\theta = 2\theta$, it becomes

$$\Delta\theta = \frac{\pi}{2}$$

at $x = 1$. At this point, the real component x and the magnitude of the imaginary component 1 are balanced. Thus, the characteristic scale R appears as a geometrical balance point in the complex

x -plane. This interpretation clarifies why the characteristic scale $r = R$ must remain explicitly encoded in the scalar-field structure.

With this background profile, the scalar field serving as a *shape function* for the spatial profile of the scale-dependent geometric repulsive response is defined by its radial derivative:

$$\phi(r) = \frac{d\mathcal{B}}{dx} \frac{dx}{dr} = \frac{d}{dr} \ln \left(1 + \frac{r^2}{R^2} \right) = \frac{2r}{R^2 + r^2}.$$

Then, $\phi(r)^2 = \frac{4}{R^2} \cdot \frac{x^2}{(1+x^2)^2}$, $\phi(r)^4 = \frac{4}{R^2} \cdot \frac{4}{R^2} \cdot \left(\frac{x^2}{(1+x^2)^2} \right)^2$, and therefore, in the standard $-\phi^2 + \phi^4$ -type potential that guarantees the symmetry $\phi \leftrightarrow -\phi$, identifying the ϕ^4 term as a higher-order term belonging to the same R -hierarchy (with a single overall prefactor of $\frac{4}{R^2}$) as the ϕ^2 term fixes the coefficient to $\frac{1}{4}R^2$. Accordingly, the potential corresponding to the scalar field $\phi(r)$ takes the form

$$\begin{aligned} V(\phi(r)) &= -\phi^2 + \frac{1}{4}R^2\phi^4 = -\left(\frac{2r}{R^2 + r^2} \right)^2 + \frac{1}{4}R^2 \left(\frac{2r}{R^2 + r^2} \right)^4 \\ &= \underbrace{-\frac{4r^2}{(R^2 + r^2)^2}}_{\text{Leading term}} + \underbrace{\frac{4R^2r^4}{(R^2 + r^2)^4}}_{\text{Second-order correction term}}. \end{aligned}$$

Here, the second term serves as a correction that slightly modifies the primary structure, but the leading term clearly satisfies the relation:

$$V(\phi(r)) \propto -\Lambda_{\text{eff}}(r), \quad \Lambda_{\text{eff}}(r) \equiv \frac{4r^2}{(R^2 + r^2)^2}.$$

In particular, in the regime $r \ll R$, the correction term is significantly smaller than the leading term. In the opposite limit $r \gg R$, their asymptotic behaviors are

$$\underbrace{\frac{r^2}{(R^2 + r^2)^2}}_{\text{Leading term}} \sim \frac{1}{r^2}, \quad \underbrace{\frac{r^4}{(R^2 + r^2)^4}}_{\text{Second-order correction term}} \sim \frac{1}{r^4},$$

indicating that the contribution from the correction term decays rapidly. Furthermore, even at the intermediate scale $r = R$, we find:

$$\underbrace{\frac{4R^2}{(2R^2)^2}}_{\text{Leading term}} = \frac{1}{R^2}, \quad \underbrace{\frac{4R^2R^4}{(2R^2)^4}}_{\text{Second-order correction term}} = \frac{1}{4R^2}.$$

showing that the correction term remains only one-fourth of the leading term.

Therefore, across all spatial scales, the leading term dominates, and it is demonstrated that $\Lambda_{\text{eff}}(r)$ can be directly and naturally derived from the scalar field potential $V(\phi(r))$.

5 Variational Evaluation of the Scalar Field

In this section, we examine whether the Lagrangian of the scalar field $\phi(r)$ —which gives rise to the geometric effective cosmological term $\Lambda_{\text{eff}}(r)$ derived in Section 4—satisfies the Euler–Lagrange extremum condition. In particular, we expect that the extremum condition is locally violated around the characteristic spatial scale $r = R$, and that this violation rapidly diminishes for $r \gg R$. Such behavior is naturally anticipated from the structure of the adopted $\phi(r)$, as it corresponds both to the notion that cosmic acceleration may be closely related to a local violation of the extremum condition and serves as a confirmation that the characteristic scale R is not lost within the structure.

Note that the violation of the extremum condition discussed here does not directly imply a violation of the conservation law $\nabla^\mu T_{\mu\nu} = 0$.

From Section 4, the scalar field and its corresponding potential take the form

$$\phi(r) = \frac{2r}{R^2 + r^2}, \quad V(\phi(r)) = -\phi^2 + \frac{1}{4}R^2\phi^4.$$

For a spherically symmetric and static configuration, the Lagrangian is given by:

$$\mathcal{L}_\phi = -\frac{1}{2} \left(\frac{d\phi}{dr} \right)^2 - V(\phi(r)) = -\frac{1}{2} \left(\frac{d\phi}{dr} \right)^2 + \phi^2 - \frac{1}{4}R^2\phi^4.$$

Thus, the Euler–Lagrange equation derived from this Lagrangian is

$$\frac{d^2\phi}{dr^2} = R^2\phi^3 - 2\phi.$$

Therefore, the deviation from the extremum condition (the variational residual) is defined as follows:

$$\delta(r) \equiv \frac{d^2\phi}{dr^2} - (R^2\phi^3 - 2\phi).$$

By substituting the adopted scalar field $\phi(r)$ into this expression, the variational residual is explicitly given by:

$$\delta(r) = \left[\frac{4r(r^2 - 3R^2)}{(R^2 + r^2)^3} \right] - \left[\frac{8R^2r^3}{(R^2 + r^2)^3} - \frac{4r}{R^2 + r^2} \right].$$

This can be rearranged as:

$$\delta(r) = \frac{4r(r^4 + r^2 + R^4 - 3R^2)}{(R^2 + r^2)^3}.$$

Consequently, the characteristic scale at which the linear term in the numerator vanishes near the origin is $R = \sqrt{3}$. At this scale, the variational residual $\delta(r)$ for the scalar field $\phi(r)$ is numerically evaluated, and the resulting profile is shown in Fig. 1.

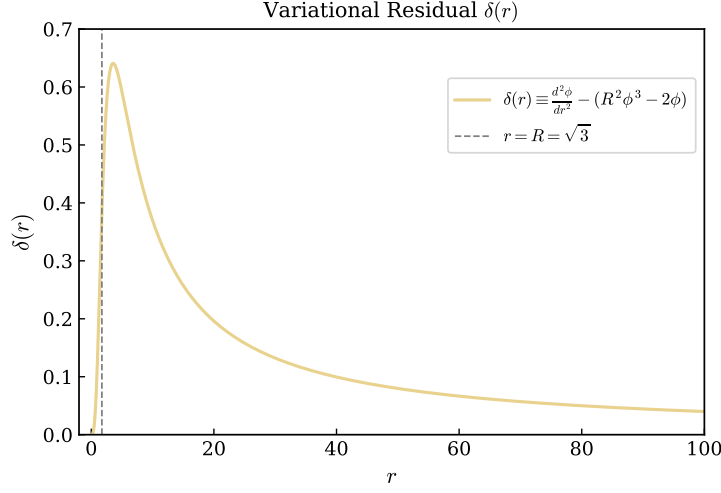


Fig. 1: Variational residual $\delta(r)$ for the scalar field $\phi(r)$.

From Fig. 1, it can be seen that the variational residual $\delta(r)$ begins to locally deviate at the spatial scale $r = R$, reaches a peak shortly thereafter ($r > R$), and then rapidly decays. This result indicates that the scalar field specifically recognizes and responds to $r = R$, and that the violation of the extremum condition is local and non-instantaneous. Such behavior is also consistent with the nature of the sustained cosmic acceleration.

The analysis is performed over the range $r \in [0.0001, 100]$. Within this interval, the maximum value of the variational residual $\delta(r)$ is found to be:

$$\delta_{\max} \approx 0.641 \quad \text{at} \quad r \approx 3.56.$$

To evaluate the significance of this residual, we define the relative error with respect to the extremum condition as:

$$\epsilon_{\phi}(r) \equiv \left| \frac{\delta(r)}{\phi(r)} \right|.$$

At the point where the residual peaks ($r \approx 3.56$), the scalar field takes the value $\phi(r) \approx 0.454$, yielding:

$$\epsilon_{\phi} \approx 1.41.$$

Similarly, at the characteristic scale $r = R = \sqrt{3}$, the relative error is:

$$\epsilon_{\phi}(R) = \left| \frac{\delta(R)}{\phi(R)} \right| \approx 0.667 \quad (\delta(R) \approx 0.385, \phi(R) \approx 0.577).$$

Even around the characteristic scale $r = R$, the relative error remains within a range that can be regarded as a local violation of the extremum condition. The value $\epsilon_{\phi}(R) \approx 0.667$ supports the validity of the function as a quasi-solution.

On the other hand, the variational residual $\delta(r)$ reaches its maximum around $r \approx 3.56$, where the relative error rises to approximately $\epsilon_{\phi} \approx 1.41$. However, since $\phi(r)$ becomes small at this scale, the relative error is numerically overestimated due to the small denominator. Therefore, it is more appropriate to evaluate the validity of the quasi-solution primarily around $r = R$.

Based on these considerations, the scalar field $\phi(r)$ is formulated as a quasi-solution so as to

preserve the characteristic scale R —encoded in the radial one-dimensional structure—rather than as an exact solution that would eliminate it.

In this context, the local violation of the variational extremum condition serves as a necessary condition for the excitation of the scalar field $\phi(r)$.

In this paper, we refer to the structure based on the effective cosmological term $\Lambda_{\text{eff}}(r)$, constructed from the quasi-solution $\phi(r)$, as the Λ_{eff} *model* (or the $f\Lambda$ *theory*).

6 Consistency with the Observed Λ

The derived scale-dependent effective cosmological term,

$$\Lambda_{\text{eff}}(r) = \frac{4r^2}{(R^2 + r^2)^2},$$

attains its maximum at $r = R$, with the value given by

$$\Lambda_{\text{eff}}^{\text{max}} = \Lambda_{\text{eff}}(r)|_{r=R} = \frac{4R^2}{(R^2 + R^2)^2} = \frac{4R^2}{4R^4} = \frac{1}{R^2}.$$

If this maximum is identified with the observed cosmological constant,

$$\Lambda \approx 10^{-52} \text{ m}^{-2},$$

then it follows that

$$R = \frac{1}{\sqrt{\Lambda}} \approx 10^{26} \text{ m},$$

which coincides with the cosmological scale, $\sim 10^{26} \text{ m}$. Therefore, it is natural to identify this characteristic scale as the effective cosmological scale, $R = \frac{1}{\sqrt{\Lambda}} \equiv R_c$.

This result implies that the effective cosmological term Λ_{eff} —derived from the scalar-field structure proposed in this study—is not only naturally consistent with the observed value of the cosmological constant but also provides a physically consistent origin for cosmic repulsion.

7 Extended Einstein Field Equation

The extended Einstein field equation is written as:

$$\Lambda_{\text{eff}}(r)g_{\mu\nu} + G_{\mu\nu} = \kappa T_{\mu\nu}^{(\text{matter})}, \quad \kappa \equiv \frac{8\pi G}{c^4}.$$

In the local regime where $r \ll R_c$ (e.g., at the solar-system scale), the geometrical term $\Lambda_{\text{eff}}(r)g_{\mu\nu}$ becomes negligibly small, and the field equation naturally reduces to the standard Einstein form:

$$G_{\mu\nu} = \kappa T_{\mu\nu}^{(\text{matter})}.$$

Therefore, this extended field equation remains consistent with well-established observations precisely explained by general relativity, such as the perihelion precession of Mercury and the gravitational redshift.

Here, the effective term $\Lambda_{\text{eff}}(r) = \frac{4r^2}{(R^2 + r^2)^2}$ can be rewritten as:

$$\Lambda_{\text{eff}}(r) = \frac{1}{R^2} F(x), \quad F(x) = \frac{4x^2}{(1+x^2)^2}, \quad x = \frac{r}{R}.$$

By applying the Bianchi identity to the extended Einstein field equation, the exchange current J —representing geometry–matter energy transfer (i.e., local non-conservation)—is defined as follows, and in a static, spherically symmetric configuration, only the radial component remains:

$$J_\nu \equiv \nabla^\mu T_{\mu\nu}^{(\text{matter})} = \frac{1}{\kappa} \partial_\nu \Lambda_{\text{eff}}, \quad J_r = \frac{1}{\kappa R^3} \frac{dF}{dx}.$$

Although local conservation is violated, both the radial and total fluxes vanish:

$$\int_0^\infty J_r dr = 0, \quad \Phi(r) = 4\pi r^2 J_r \Rightarrow \Phi(0) = \Phi(\infty) = 0.$$

Thus, the geometry–matter energy transfer is exactly balanced throughout the volume, ensuring global energy conservation.

Additionally, at cosmological scales $r \sim R_c \sim 10^{26}$ m, where the local non-conservation represented by J_r becomes negligibly small,

$$\left| \frac{d\Lambda_{\text{eff}}}{dr} \right| \sim 10^{-78} \text{ m}^{-3} \Rightarrow \Lambda_{\text{eff}} \approx \text{const.},$$

and hence the Friedmann equations are naturally recovered:

$$H^2 \equiv \left(\frac{\dot{a}}{a} \right)^2 = \frac{8\pi G}{3} \rho - \frac{kc^2}{a^2} + \frac{\Lambda c^2}{3}, \quad \frac{\ddot{a}}{a} = -\frac{4\pi G}{3} \left(\rho + \frac{3P}{c^2} \right) + \frac{\Lambda c^2}{3}.$$

8 Application to Galactic Rotation Curves

8.1 Gravitational Term: Realistic Linear Approximation

In the central region of a galaxy, the baryonic matter is relatively concentrated, allowing the use of a spherically symmetric and uniform density approximation. Under this assumption, the mass $M(r)$ enclosed within a radius r is given by:

$$\rho = \text{const.} \Rightarrow M(r) = \frac{4}{3}\pi r^3 \rho.$$

From the gravitational potential generated by this mass distribution, the rotational velocity is derived as:

$$v_{\text{grav}}^2(r) = \frac{GM(r)}{r} = \frac{G}{r} \left(\frac{4}{3}\pi r^3 \rho \right) = \frac{4}{3}\pi G \rho r^2, \quad \Rightarrow \quad v_{\text{grav}} \propto r.$$

The relation $v_{\text{grav}} \propto r$ agrees with the observed linear rise in the inner regions of galaxies.

8.2 Repulsive Term: Rotational Velocity and Flattening

While the gravitational term from the uniform density model dominates in the central region of galaxies, explaining the flattened rotation curves in the outer regions requires a different dynamical contribution.

In this context, a scale-dependent effective Λ term, generated by the spatial configuration of the scalar field $\phi(r)$, yields a geometric repulsive effect that contributes to the squared rotational velocity as:

$$v_{\text{rep}}^2(r) = v_0^2 \cdot \frac{2r^2}{R^2 + r^2},$$

where:

- v_0 is a characteristic velocity scale (unit: m/s) associated with the repulsive term,
- which becomes dominant in the outer galactic region ($r > R$).

Since this function contributes negligibly near the center and asymptotically approaches a constant value $v_{\text{rep}} \rightarrow \sqrt{2} v_0$ as $r \rightarrow \infty$, it naturally reproduces the observed flattening of galactic rotation curves. Based on observational data, we adopt a representative value of $v_0 \approx 200$ km/s.

8.3 Rotational Velocity from Energy Composition

In this model, the gravitational and repulsive contributions to the rotational velocity are regarded as independent energy components. Therefore, the observed rotational velocity is expressed as a quadratic energy sum:

$$v_{\text{total}}^2(r) = v_{\text{grav}}^2(r) + v_{\text{rep}}^2(r) \quad \Rightarrow \quad v_{\text{total}}(r) = \sqrt{v_{\text{grav}}^2(r) + v_{\text{rep}}^2(r)}.$$

This composite structure implies that:

- in the inner region, $v_{\text{grav}} \propto r$ dominates,
- while in the outer region, $v_{\text{rep}} \rightarrow \sqrt{2} v_0$ flattens the rotation curve.

Overall, this framework demonstrates that the essential features of galactic rotation curves can be reproduced without assuming the existence of dark matter, thereby supporting the validity of the effective Λ term, Λ_{eff} .

8.4 Velocity $v_0 \approx 200$ km/s from the Effective Energy of Λ_{eff}

Based on the result of Section 6 and the vacuum-energy density expression in the Λ CDM model, we obtain the *geometric effective energy density*:

$$\rho_{\Lambda_{\text{eff}}}(R_c) = \rho_{\Lambda} = \frac{\Lambda c^2}{8\pi G} \approx 10^{-26} \text{ kg/m}^3.$$

By integrating this geometric effective energy density over a spherically symmetric region of radius $\sim 1000 \text{ kpc} \approx 3.1 \times 10^{22} \text{ m}$, the corresponding geometric effective mass can be obtained. Since $10^{22} \text{ m} \ll 10^{26} \text{ m}$, the effective energy density $\rho_{\Lambda_{\text{eff}}}(R_c)$ can therefore be regarded as nearly constant within this scale range, yielding:

$$M_{\Lambda} \approx \rho_{\Lambda} \cdot \frac{4}{3}\pi r^3 \approx 10^{42} \text{ kg}.$$

If we convert a part of the effective mass energy into kinetic energy, it follows that:

$$E_{\text{rot}} = \frac{1}{2}M_{\Lambda}v_0^2 \quad \Rightarrow \quad v_0 \approx 200 \text{ km/s} \approx \sqrt{\frac{2E_{\text{rot}}}{M_{\Lambda}}}.$$

Therefore, the characteristic velocity $v_0 \approx 200 \text{ km/s}$ can be naturally estimated within the framework of the Λ_{eff} model, and the corresponding energy $E_{\text{rot}} \approx 2 \times 10^{52} \text{ J}$ is consistent with the typical observational energy scale $E_{\text{obs}} \sim 10^{52} \text{ J}$. Considering that the total geometric effective energy is given by $E_{\Lambda} = M_{\Lambda}c^2 \approx 10^{59} \text{ J}$, the magnitude of E_{rot} is sufficiently small in comparison, ensuring energetic consistency.

These results quantitatively support the physical plausibility of the scale-dependent term Λ_{eff} , derived from the scalar field, as an effective cosmological repulsive component.

8.5 Physical Interpretation: Dominant Structures by Region

- **Central region** ($r < R$):

The gravitational term dominates, with $v_{\text{grav}} \propto r$.

- **Intermediate region** ($r \approx R$):

The repulsive term v_{rep} begins to contribute, marking the transition toward velocity flattening.

- **Outer region** ($r > R$):

The repulsive contribution dominates, and the rotation velocity becomes asymptotically flat, reflecting the logarithmic-type structure and scale dependence of Λ_{eff} .

Representative Parameters and Plotting Conditions

- Gravitational constant: $G = 4.302 \times 10^{-6} \text{ kpc} \cdot (\text{km/s})^2 / M_{\odot}$
- Mass density (average over the galaxy): $\rho = 10^6 M_{\odot} / \text{kpc}^3$
- Characteristic scale of the galaxy: $R = 5 \text{ kpc}$
- Characteristic velocity of the galaxy: $v_0 = 200 \text{ km/s}$
- Radial range for calculation: $r = 0.1 \sim 20 \text{ kpc}$

Using the above representative values, the numerically plotted rotation velocity profile at the galactic scale is shown in Fig. 2, based on the previously derived composite formula:

$$v_{\text{total}}(r) = \sqrt{v_{\text{grav}}^2(r) + v_{\text{rep}}^2(r)}.$$

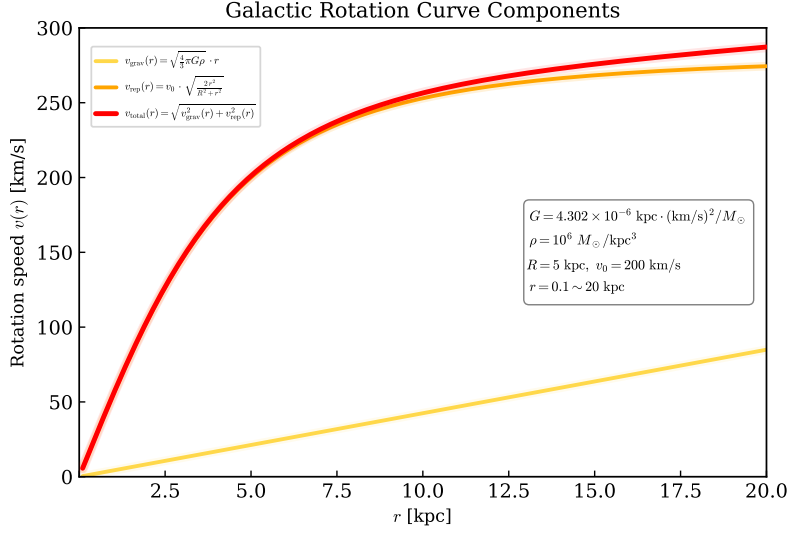


Fig. 2: Numerical simulation of gravitational, repulsive, and total rotational velocity components for a galactic rotation curve.

As clearly demonstrated by the numerical simulation results shown in Fig. 2, it was initially expected that the gravitational term would dominate in the central region ($r < R$). However, the repulsive component was quantitatively found to exceed the gravitational component at all radii within the galaxies modeled. This indicates that the rotational velocity within galaxies is effectively governed by the scalar-field-induced effective Λ term—characterized by its logarithmic-type structure and defined by the characteristic galactic scale $R \sim \text{kpc} \equiv R_g$.

This result not only explains the observed flattening of rotational velocities in the outer regions of galaxies, but also reveals that the repulsive term significantly contributes even in the central region. It strongly indicates that the scale-dominant behavior of the logarithmic-type repulsive term extends far more broadly than previously anticipated.

Therefore, the galactic rotation curves are systematically reproduced by the repulsive component:

$$v_{\text{rep}}(r) = v_0 \cdot \sqrt{\frac{2r^2}{R^2 + r^2}}.$$

9 BH Singularity Avoidance and the Critical Radius

The repulsive acceleration derived from the scalar field $\phi(r)$ is given by:

$$a_{\text{rep}}(r) = c_0 \cdot \phi(r) = \frac{2c_0 r}{R^2 + r^2}.$$

Here, c_0 is a repulsive potential coefficient (unit: m^2/s^2), representing the characteristic spatial energy scale of the repulsive field. Note that this acceleration formula corresponds uniquely to the rotational velocity function defined in Section 8.2 through the circular motion condition $a = \frac{v^2}{r}$:

$$a_{\text{rep}}(r) = \frac{2c_0 r}{R^2 + r^2} \quad \begin{matrix} c_0 = v_0^2 \\ \longleftrightarrow \\ a = \frac{v^2}{r} \end{matrix} \quad v_{\text{rep}}^2(r) = \frac{2v_0^2 r^2}{R^2 + r^2} = v_0^2 \cdot \frac{2r^2}{R^2 + r^2}.$$

Inside the black hole, the repulsive acceleration is considered to balance the gravitational acceleration. Under this condition, the balance between the repulsive and gravitational forces is expressed as:

$$a_{\text{rep}}(r) = g(r) \Rightarrow \frac{2c_0 r}{R^2 + r^2} = \frac{GM}{r^2}.$$

Here, M denotes the mass of the black hole.

At $r = R = r_{\text{crit}}$, where $a_{\text{rep}}(r)$ attains its maximum $a_{\text{rep}}^{\text{max}}$, the balance condition between the repulsive and gravitational accelerations is expressed as:

$$\frac{c_0}{r_{\text{crit}}} = \frac{GM}{r_{\text{crit}}^2} \Rightarrow c_0 = \frac{GM}{r_{\text{crit}}}.$$

Here, the repulsive energy at the critical radius is given by:

$$E_{\text{rep}} = M \cdot c_0 = \frac{GM^2}{r_{\text{crit}}},$$

while the total mass-energy of the black hole is

$$E_{\text{BH}} = Mc^2.$$

Within this setup—where the repulsive response of space is induced by the black hole’s own mass-energy—the system can be treated as a spherically symmetric and static closed configuration. By the principle of energy conservation, the repulsive energy accumulated within the black hole cannot exceed its total mass-energy. Accordingly, equating the two energies defines the critical radius at which further collapse becomes energetically forbidden:

$$E_{\text{rep}} = E_{\text{BH}} \Rightarrow \frac{GM^2}{r_{\text{crit}}} = Mc^2 \Rightarrow r_{\text{crit}} = \frac{GM}{c^2}.$$

This critical radius corresponds exactly to half of the Schwarzschild radius:

$$r_{\text{crit}} = \frac{1}{2} r_s, \quad \text{where} \quad r_s = \frac{2GM}{c^2}.$$

Thus, the characteristic scale of a black hole is naturally defined as $R = \frac{1}{2} r_s \equiv R_{\text{BH}}$.

Since the condition $M \cdot c_0 = Mc^2$ ($E_{\text{rep}} = E_{\text{BH}}$) is satisfied at the critical radius, it follows that $c_0 = c^2$ at that point. **Consequently, singularities are prohibited by the principle of energy conservation and the speed-of-light limit.** The singularity is therefore avoided not by introducing an external cutoff, but by the internal repulsive structure of space itself.

As a direct geometric consequence, the apparent black hole shadow is characterized by the critical radius $r_{\text{crit}} = \frac{GM}{c^2}$. The photon sphere is located at

$$r_{\text{ph}} = \frac{3}{2} r_s = 3 r_{\text{crit}},$$

and the apparent shadow radius, including gravitational lensing, is

$$b_c = \frac{r_{\text{ph}}}{\sqrt{1 - \frac{2r_{\text{crit}}}{r_{\text{ph}}}}} = 3\sqrt{3}r_{\text{crit}}.$$

Hence, the shadow diameter becomes

$$D = 2b_c = 6\sqrt{3}r_{\text{crit}},$$

yielding a dimensionless ratio:

$$\frac{D}{r_{\text{crit}}} = 6\sqrt{3} \approx 10.4,$$

which is universally applicable to any black hole without normalization and agrees remarkably well with the Event Horizon Telescope (EHT) measurements, M87* $d/\theta_g \approx 11.1$ [6] and Sgr A* $d/\theta_g \approx 10.8$ [7].

With $a_{\text{rep}}^{\text{max}}$ the repulsive acceleration at r_{crit} , and since $c_0 = c^2$ there, we have:

$$a_{\text{rep}}^{\text{max}} = \frac{c^2}{r_{\text{crit}}}.$$

Substituting $r_{\text{crit}} = \frac{GM}{c^2}$ gives:

$$a_{\text{rep}}^{\text{max}} = \frac{c^2}{\frac{GM}{c^2}} = \frac{c^4}{GM}.$$

Therefore, both the repulsive acceleration and the gravitational acceleration inside a black hole cannot exceed this value.

Here, we take $a_{\text{rep}}^{\text{max}}$ to be equal to the Planck acceleration $a_{\text{P}} \equiv \sqrt{\frac{c^7}{\hbar G}}$, and we obtain:

$$\frac{c^4}{GM} = \sqrt{\frac{c^7}{\hbar G}} \implies M = \sqrt{\frac{\hbar c}{G}}.$$

This value corresponds to the Planck mass $m_{\text{P}} \equiv \sqrt{\frac{\hbar c}{G}}$, showing that the mass of a black hole, even when compressed to the Planck scale, cannot fall below this fundamental limit.

10 Consistency with the Higgs Mechanism

Following the foundational standpoint of Section 3, matter and four-dimensional spacetime are regarded in the present framework as secondary entities generated from the one-dimensional geometrical repulsive response structure. Therefore, the origin of mass must be identified as a manifestation of the same scalar-field structure, without being introduced as an external postulate. Accordingly, this section clarifies that the scalar potential of the Λ_{eff} model defined in Section 4,

$$V(\phi(r)) = -\phi^2 + \frac{1}{4}R^2\phi^4,$$

is directly embedded in the mass-generating structure of the Higgs mechanism.

To obtain the vacuum expectation value (VEV), we differentiate the potential $V(\phi(r))$ in the Λ_{eff} model:

$$\frac{dV(\phi(r))}{d\phi} = 0 \quad \Rightarrow \quad \phi(-2 + R^2\phi^2) = 0.$$

Therefore, the solutions are

$$\phi = 0 \quad \text{or} \quad \phi = \pm \frac{\sqrt{2}}{R}.$$

Thus, excluding the trivial solution $\phi = 0$, the vacuum expectation value in the Λ_{eff} model is

$$v = \frac{\sqrt{2}}{R}, \quad \phi = \pm v.$$

From this relation, the characteristic scale R can be expressed in terms of the experimentally determined Higgs vacuum expectation value $v \approx 246 \text{ GeV}$ as

$$R = \frac{\sqrt{2}}{v} = \frac{\sqrt{2}}{246 \text{ GeV}} \approx 5.75 \times 10^{-3} \text{ GeV}^{-1}.$$

Using the natural unit conversion ($\hbar = c = 1$) with $\text{GeV}^{-1} \approx 1.97 \times 10^{-16} \text{ m}$, we obtain

$$R \approx 5.75 \times 10^{-3} \times 1.97 \times 10^{-16} \text{ m} \approx 1.13 \times 10^{-18} \text{ m}.$$

This value precisely corresponds to the electroweak scale at which the Higgs boson was discovered in experiments at the Large Hadron Collider (LHC), at $\sim 10^{-18} \text{ m}$. Therefore, it is natural to identify this characteristic scale as the Higgs scale, $R = \frac{\sqrt{2}}{v} \equiv R_h$, and the Yukawa coupling can be defined through a geometric relation as

$$y_f = \frac{R_h}{R_f}, \quad R_f \equiv \frac{1}{m_f},$$

or, taking logarithms, as a distance in log-scale space:

$$-\ln y_f = \Delta\nu, \quad \Delta\nu \equiv \ln R_f - \ln R_h.$$

Here, the potential for the Higgs field in the Standard Model is given by:

$$V_{\text{Higgs}}(\phi) = -\frac{1}{2}\mu^2\phi^2 + \frac{1}{4}\lambda\phi^4, \quad v^2 = \frac{\mu^2}{\lambda}, \quad m_h^2 = 2\lambda v^2.$$

By substituting these relations, the potential can be rewritten as:

$$V_{\text{Higgs}}(\phi) = -\frac{1}{4}m_h^2\phi^2 + \frac{m_h^2}{8v^2}\phi^4.$$

Thus, the Higgs potential can be expressed in a subdivided form:

$$V_{\text{Higgs}}(\phi) = -\left(\frac{m_h}{2}\right)^2\phi^2 + \frac{1}{4}\left(\frac{\sqrt{2}}{v}\right)^2\left(\frac{m_h}{2}\right)^2\phi^4.$$

Here, let us define:

$$\frac{m_h}{2} \equiv \check{m},$$

and substitute the previously obtained relation $R_h = \frac{\sqrt{2}}{v}$:

$$V_{\text{Higgs}}(\phi) = -\check{m}^2 \phi^2 + \frac{1}{4} R_h^2 \check{m}^2 \phi^4.$$

Rearranging gives

$$V_{\text{Higgs}}(\phi) = \check{m}^2 \left(-\phi^2 + \frac{1}{4} R_h^2 \phi^4 \right).$$

This suggests the natural choice

$$\check{m} = 1$$

to bridge mass-dimensional and dimensionless forms. By adopting $\hbar = c = \check{m} = 1$ and replacing $R_h = \frac{\sqrt{2}}{v}$ with a general characteristic scale R , we obtain a universal scalar potential in a mass-dimensionless form:

$$V_{\text{Higgs}}(\phi) = -\phi^2 + \frac{1}{4} R^2 \phi^4$$

which can be applied to general scales and is identical to the potential of the Λ_{eff} model $V(\phi(r)) = -\phi^2 + \frac{1}{4} R^2 \phi^4$.

By assigning $\check{m}$ as the characteristic mass in $V(\phi(r))$ at the Higgs scale, the self-coupling constant is obtained as:

$$\lambda_{\Lambda_{\text{eff}}} \equiv R_h^2 \check{m}^2.$$

In natural units ($\hbar = c = 1$), where $\check{m} \neq 1$ is explicitly retained as the mass scale, using $R_h \approx 1.13 \times 10^{-18} \text{ m} \approx 5.75 \times 10^{-3} \text{ GeV}^{-1}$ and $\check{m} = \frac{m_h}{2} \approx 62.5 \text{ GeV}$, we obtain

$$\lambda_{\Lambda_{\text{eff}}} \approx (5.75 \times 10^{-3} \text{ GeV}^{-1})^2 \cdot (62.5 \text{ GeV})^2 \approx 0.129,$$

which coincides with the Standard Model Higgs self-coupling constant, $\lambda_{\text{SM}} \approx 0.129$.

These results in this section indicate that mass generation is driven by the geometric repulsive response.

Is Double-Higgs (hh) Production Possible?

The production of a single Higgs boson (h) occurs when two half-mass quantum states $|m_h/2\rangle_{1,2}$ coherently superpose within the squared region of the characteristic R_h -scale:

$$\frac{1}{\sqrt{2}} \left(|m_h/2\rangle_1 + |m_h/2\rangle_2 \right) \Rightarrow |h\rangle.$$

This coherent-overlap condition is reflected in the relation

$$\lambda_{\Lambda_{\text{eff}}} = R_h^2 \left(\frac{m_h}{2} \right)^2.$$

In contrast, the production of a double-Higgs (hh) consists of two independent single-Higgs superposition states,

$$|hh\rangle = |h\rangle \otimes |h\rangle = \left(\frac{1}{\sqrt{2}}\right)^2 \left(|m_h/2\rangle_1 + |m_h/2\rangle_2\right) \otimes \left(|m_h/2\rangle_3 + |m_h/2\rangle_4\right).$$

However, since the characteristic length scale $R_h \approx 1.13 \times 10^{-18}$ m corresponds to an energy scale $1/R_h \approx 174$ GeV, localizing four half-mass components with total energy ≈ 250 GeV within the same R_h -scale region would exceed this limit:

$$E_{hh} \equiv 4 \cdot \left(\frac{m_h}{2}\right) \approx 250 \text{ GeV} > E_{R_h} \equiv \frac{1}{R_h} \approx 174 \text{ GeV}.$$

Consequently, the four-point overlap — i.e., double-Higgs production — is energetically prohibited by the R_h -scale confinement limit.

11 Discussion and Conclusion

In this study, we derived the following scale-dependent effective cosmological term from the scalar field $\phi(r)$, which is formulated as a quasi-solution that preserves the characteristic scale R while remaining consistent with the variational principle:

$$\Lambda_{\text{eff}}(r) = \frac{4r^2}{(R^2 + r^2)^2}.$$

This term functions effectively in the following three aspects:

1. Consistency with cosmology:

The effective term Λ_{eff} reaches its maximum value at $r = R$, where

$$\Lambda_{\text{eff}}^{\text{max}} = \frac{1}{R^2}.$$

By identifying this with the observed cosmological constant, we obtain $R \approx 10^{26}$ m, which coincides with the cosmological scale, $\sim 10^{26}$ m.

2. Reproduction of galactic rotation curves:

Using the repulsive term based on Λ_{eff} , we constructed the following composite formula for rotational velocity:

$$v_{\text{total}}(r) = \sqrt{v_{\text{grav}}^2(r) + v_{\text{rep}}^2(r)}, \quad v_{\text{grav}}^2(r) = \frac{4}{3}\pi G\rho r^2, \quad v_{\text{rep}}^2(r) = v_0^2 \cdot \frac{2r^2}{R^2 + r^2}.$$

Numerical evaluation under a realistic galactic-scale configuration and representative parameters confirmed that this structure successfully reproduces the observed flat rotation velocity profiles of galaxies. As a result, the geometric repulsive term associated with Λ_{eff} was found to be dominant over the gravitational contribution throughout the galaxy.

Therefore, the galactic rotation curves are systematically characterized by the repulsive component:

$$v_{\text{rep}}(r) = v_0 \cdot \sqrt{\frac{2r^2}{R^2 + r^2}}.$$

3. Critical radius and singularity avoidance

By deriving the condition under which the repulsive energy equals the mass–energy of the black hole, we obtained a critical radius $r_{\text{crit}} = \frac{1}{2}r_s$ at which collapse is doubly locked by energy conservation and the speed-of-light limit. This indicates that singularities can be naturally avoided within the internal dynamics of the geometric structure itself, without requiring any external cutoff.

Consequently, the contribution of Λ_{eff} manifests as a geometrical effect opposite in direction to conventional gravity, playing a unifying role across scales—from the accelerated expansion of the universe and the flattening of galactic rotation curves to the prevention of black hole singularities.

Continued theoretical advancement of the model, along with its observational and empirical verification, remains a central task. As a first step in this direction, we have confirmed its consistency with the Higgs mechanism within the scalar potential framework of the f Λ theory.

Acknowledgements

This research was solely developed by the author, without external collaboration.

Funding Statement

No external funding was received for this work.

Ethical Statement

This study involved no human or animal subjects; ethical approval was not applicable.

Conflict of Interest Statement

The author declares no conflict of interest.

References

- [1] V. C. Rubin and W. K. Ford, Jr. Rotation of the Andromeda Nebula from a Spectroscopic Survey of Emission Regions. *The Astrophysical Journal*, 159:379–403, 1970.
 - [2] Y. Sofue and V. Rubin. Rotation Curves of Spiral Galaxies. *Annual Review of Astronomy and Astrophysics*, 39:137–174, 2001.
 - [3] Roger Penrose. Gravitational Collapse and Space-Time Singularities. *Physical Review Letters*, 14(3):57–59, 1965.
 - [4] Planck Collaboration. Planck 2018 results. VI. Cosmological parameters. *Astronomy & Astrophysics*, 641:A6, 2020.
 - [5] T. Padmanabhan. Cosmological Constant - The Weight of the Vacuum. *Physics Reports*, 380:235–320, 2003.
 - [6] Event Horizon Telescope Collaboration. First M87 Event Horizon Telescope Results. VI. The Shadow and Mass of the Central Black Hole. *The Astrophysical Journal Letters*, 875(1):L6, 2019.
 - [7] Event Horizon Telescope Collaboration. First Sagittarius A* Event Horizon Telescope Results. IV. Variability, Morphology, and Black Hole Mass. *The Astrophysical Journal Letters*, 930(2):L15, 2022.
-

Appendix A

Cosmological Mapping of the Variational Residual Structure

In Section 3, the geometric scale ratio $\frac{r}{R}$ was positioned as the basis of time evolution. On this basis, this appendix summarizes how $\Lambda_{\text{eff}}(r)$ and the variational residual structure in the present theory (f Λ) are mapped onto cosmology.

From Section 5, the variational residual at $R = \sqrt{3}$ is

$$\delta(r) = \frac{4r^3(r^2 + 1)}{(r^2 + 3)^3}$$

and it takes its maximum value $\delta_{\text{max}} \simeq 0.641$ at $r_{\text{peak}} \simeq 3.56$. As noted in Section 4, since both r and R are treated as dimensionless quantities, r_{peak} is also dimensionless.

From Section 7, the base response function of $\Lambda_{\text{eff}}(r)$ is

$$F(x) = \frac{4x^2}{(1 + x^2)^2}, \quad x = \frac{r}{R}.$$

Here, since the physical length in FLRW spacetime is given by $\ell_{\text{phys}}(t) = a(t) \ell_{\text{com}}$, we have $x(t) \equiv \frac{\ell_{\text{phys}}(t)}{R} = \frac{a(t) \ell_{\text{com}}}{R}$. Therefore, if we normalize the comoving scale as $\ell_{\text{com}} = R$ with respect to the characteristic scale R , then $x(t) = a(t)$. Accordingly, with the standard relation $a(z) = (1 + z)^{-1}$, the cosmological mapping is

$$x(z) = a(z) = (1 + z)^{-1}$$

and, by defining $\Xi_{\text{base}}(z) \equiv F(x(z))$, we obtain

$$\Xi_{\text{base}}(z) = \frac{4(1 + z)^2}{[(1 + z)^2 + 1]^2}.$$

If one assumes that the residual strength is reflected in local cosmological expansion, then from the dimensionless relation $\frac{r}{r_{\text{peak}}} = \frac{z}{z_c}$ (i.e., $r = r_{\text{peak}} \cdot \frac{z}{z_c}$), the variational residual $\delta(r)$ is expressed as

$$\delta\left(r_{\text{peak}} \cdot \frac{z}{z_c}\right).$$

Normalizing this by δ_{max} , one obtains the normalized window function

$$W_{\delta}(z; z_c) = \frac{\delta\left(r_{\text{peak}} \cdot \frac{z}{z_c}\right)}{\delta_{\text{max}}}$$

which is directly proportional to the variational residual and satisfies both $W_{\delta}(0; z_c) = 0$ and $W_{\delta}(z_c; z_c) = 1$. Therefore, the effective expansion response function is given by

$$\Xi(z) = \Xi_{\text{base}}(z) [1 + \epsilon W_{\delta}(z; z_c)]$$

and the dimensionless expansion rate and the Hubble parameter are respectively written as

$$E^2(z) = \Omega_r(1+z)^4 + \Omega_m(1+z)^3 + \Omega_\Lambda \Xi(z), \quad H(z) = H_0 E(z).$$

In this study, we assume an approximately flat spacetime background and adopt the representative values $\Omega_m(\equiv \Omega_{m0}) = 0.30$, $\Omega_r = 9 \times 10^{-5}$, $\Omega_\Lambda = 1 - \Omega_m - \Omega_r$, and $H_0 = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$. We also apply the maximum residual value δ_{max} as the normalization coefficient, namely,

$$\epsilon = \delta_{\text{max}} \simeq 0.641.$$

A.1 Effective Redshift Scale and the H_0 Tension

The quantity z_c is the effective redshift scale, and it is natural to regard its value as being determined by the distance to the cosmic microwave background (CMB) at each cosmic epoch. Therefore, to determine the value of z_c today, it is sufficient to evaluate the point at which the relative difference in the comoving distance to the last-scattering surface at $z_* = 1089.9$,

$$\Delta D_C = \frac{D_C^{\text{f}\Lambda} - D_C^{\Lambda\text{CDM}}}{D_C^{\Lambda\text{CDM}}},$$

is minimized.

Table 1: Relative difference in the comoving distance to the last-scattering surface, $\Delta D_C(z_*)$, and its dependence on z_c . The calculation was performed for $z_* = 1089.9$, using $\Omega_m = 0.30$, $\Omega_r = 9 \times 10^{-5}$, $\Omega_\Lambda = 1 - \Omega_m - \Omega_r$, $r_{\text{peak}} = 3.56$, and $\epsilon = \delta_{\text{max}} = 0.641$.

z_c	$\Delta D_C(z_* = 1089.9)$
0.020	+0.762 %
0.050	+0.304 %
0.080	+0.0204 %
0.0820	+0.00515 %
0.0827	−0.000100 %
0.0830	−0.00234 %
0.100	−0.117 %
0.150	−0.349 %
0.200	−0.483 %

From this result, we find that the relative difference in the comoving distance to the last-scattering surface is minimized, while changing sign, in the vicinity of $z_c \simeq 0.0827$. Therefore, the effective redshift scale today is determined to be

$$\boxed{z_c \simeq 0.0827}.$$

Using this value of z_c , we define H_0^{app} as the best-fit value of H_0 obtained by fitting the ΛCDM distance modulus $\mu_{\Lambda\text{CDM}}(z; H_0)$ to the distance modulus of the present model,

$$\mu_{\text{f}\Lambda}(z) = 5 \log_{10} \left(\frac{d_L^{\text{f}\Lambda}(z)}{\text{Mpc}} \right) + 25, \quad d_L^{\text{f}\Lambda}(z) = (1+z) \frac{c}{H_0} \int_0^z \frac{dz'}{E_{\text{f}\Lambda}(z')},$$

evaluated over the Hubble-flow range $0.0233 < z < 0.15$ in the unweighted least-squares sense, which yields

$$H_0^{\text{app}} \simeq 73.6 \text{ km s}^{-1} \text{ Mpc}^{-1}.$$

This means that, while retaining the early-universe Λ CDM reference value $67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, a larger apparent Hubble constant is observed in the local universe around $z_c \simeq 0.0827$, which is also consistent with the representative value $H_0 \simeq 73 \text{ km s}^{-1} \text{ Mpc}^{-1}$ in the range $0.0233 < z < 0.15$ reported by SH0ES [8].

A.2 S_8 Tension

To maintain consistency in how the variational residual is reflected in the expansion sector, we introduce the same window function W_δ into the effective gravitational coupling,

$$\mu(z) = 1 - \eta W_\delta(z; z_c),$$

with the normalization coefficient $\eta = \delta_{\text{max}} \simeq 0.641$.

With this effective coupling $\mu(z)$, the linear growth equation is given by

$$\frac{d^2 D}{da^2} + \frac{3 + \frac{d \ln H}{d \ln a}}{a} \frac{dD}{da} - \frac{3}{2a^2} \mu(z) \Omega_m(a) D = 0, \quad \Omega_m(a) = \frac{\Omega_{m0} a^{-3}}{E(a)^2}, \quad a = (1+z)^{-1},$$

which we solve numerically, yielding

$$S_8 = \sigma_8 \sqrt{\frac{\Omega_{m0}}{0.3}} \simeq 0.778.$$

This result implies a smaller amplitude of structure growth than that predicted by standard Λ CDM, and is also consistent with the value $S_8 \simeq 0.77$ inferred from weak-lensing observations such as KiDS [9]. (In this evaluation, the primordial fluctuation amplitude is taken to be the same as in Λ CDM, and σ_8 is estimated from the ratio of the growth function $D(1)$ at $a = 1$, using $\sigma_{8,\Lambda\text{CDM}} = 0.811$ as the reference value [4]. Since we adopt $\Omega_{m0} = 0.30$ in this work, we have $S_8 = \sigma_8$ in this case.)

Therefore, the variational residual structure derived from the present theory simultaneously alleviates the H_0 tension and the S_8 tension identified in standard cosmology, while preserving the geometry of the CMB, as summarized below:

$$\boxed{\begin{array}{l} z_c \simeq 0.0827, \quad \epsilon = \eta = \delta_{\text{max}} \\ H_0^{\text{app}} \simeq 73.6 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad S_8 \simeq 0.778, \quad \Delta D_C(z_* = 1089.9) \approx -0.000100 \% \end{array}}.$$

Finally, the behavior of the dimensionless expansion rate $E(z)$ in the present model, compared with that of Λ CDM, is shown in Fig. 3.

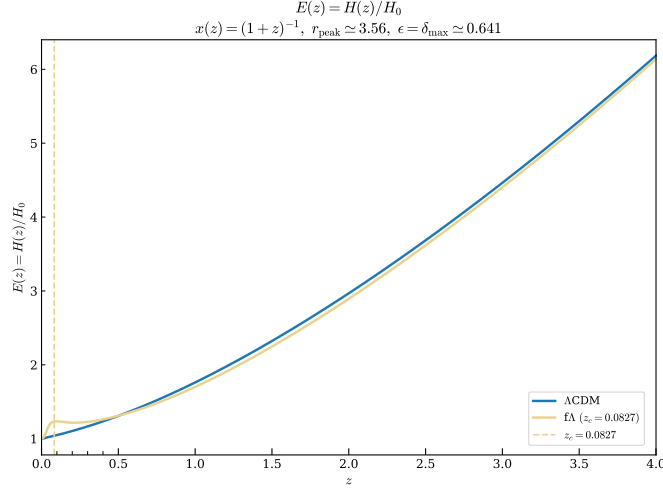


Fig. 3: Behavior of the dimensionless expansion rate $E(z)$ compared with Λ CDM.

In addition, a comparison of $f\sigma_8(z)$ with overlaid observational data points from Table I of [10], approximately corrected to the reference value $\Omega_{m0} = 0.30$, is presented in Fig. 4. Here, as in Λ CDM, $f\sigma_8(z) = f(z)\sigma_8(z)$, where the growth rate is defined by $f(z) \equiv \frac{d \ln D}{d \ln a}$.

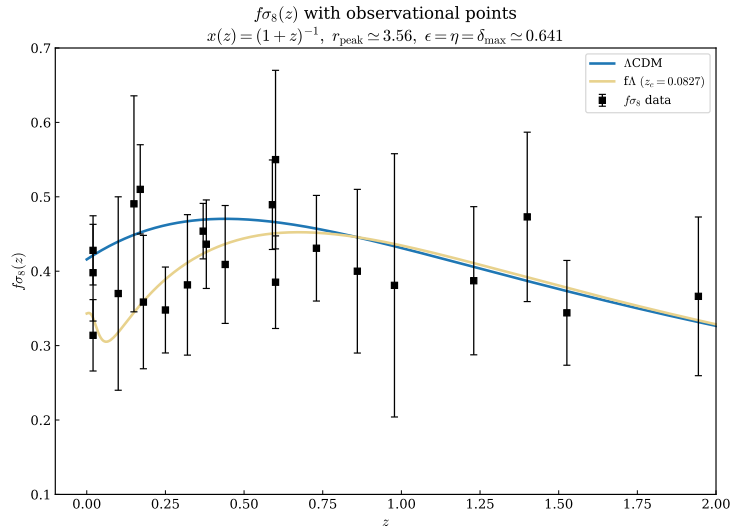


Fig. 4: Comparison of $f\sigma_8(z)$ with observational data points overlaid. The $f\sigma_8(z)$ curve of the present model (yellow) can be seen to reproduce the observed structure-growth data relatively well.

Additional References

- [8] A. G. Riess et al. A Comprehensive Measurement of the Local Value of the Hubble Constant with $1 \text{ km s}^{-1} \text{ Mpc}^{-1}$ Uncertainty from the Hubble Space Telescope and the SH0ES Team. *The Astrophysical Journal Letters*, 934:L7, 2022.
- [9] S.-S. Li et al. KiDS-1000: Cosmology with improved cosmic shear measurements. *Astronomy & Astrophysics*, 679:A133, 2023.
- [10] B. Sagredo, S. Nesseris, and D. Sapone. Internal Robustness of Growth Rate data. *Physical Review D*, 98:083543, 2018, arXiv:1806.10822 [astro-ph.CO].

B The Matter-Like Geometric Component Resisting the Variational Residual and Ω_{m0}

In this appendix, we formulate the critical restoring response that can resist the rise of the variational residual as a matter-like effective geometric component.

As shown in Section 4, the scalar field $\phi(r)$ is treated in the present theory as a dimensionless amplitude characterizing the geometric response. Under the assumption of spherical symmetry, the normalized amplitude of the local geometric response is represented by the dimensionless amplitude $\phi(r)$ normalized by the radius r , namely $\phi(r)/r$.

To describe this amplitude as an effective component in the cosmological background, it is necessary to use the cosmological length scale $R_c = 1/\sqrt{\Lambda}$, defined in Section 6, as the reference length in the denominator. That is, $\phi(r)/R_c$ represents the amplitude of the local geometric response normalized to the cosmological background scale, and its strength is characterized by $(\phi(r)/R_c)^2$. To convert this strength into the form of an effective density, we define the effective density of the geometric resistive component using the gravitational coupling constant $\kappa = 8\pi G/c^4$ as

$$\rho_{\text{geo}}^{\text{resist}}(r) \equiv \frac{1}{\kappa} \left(\frac{\phi(r)}{R_c} \right)^2.$$

Here, as discussed in Appendix A, $x(z) = a(z) = (1+z)^{-1}$ under the comoving scale normalization $\ell_{\text{com}} = R$. Therefore, from the relation $\ell_{\text{phys}} = a\ell_{\text{com}}$ at the present epoch, namely at $a = 1$, it follows that

$$r = \ell_{\text{phys}} = \ell_{\text{com}} = R.$$

Accordingly, the present value of the effective density of the geometric resistive component is given by

$$\rho_{\text{geo},0}^{\text{resist}} = \frac{1}{\kappa} \left(\frac{\phi(R)}{R_c} \right)^2.$$

Normalizing this by the present critical density $\rho_{\text{crit},0} = \frac{3H_0^2 c^2}{8\pi G}$, we obtain

$$\Omega_{\text{geo},0}^{\text{resist}} = \left(\frac{\phi(R)}{R_c} \right)^2 \frac{c^2}{3H_0^2}.$$

Moreover, since Section 6 gives $1/R_c^2 = \Lambda$, this can be reduced to

$$\Omega_{\text{geo},0}^{\text{resist}} = \phi(R)^2 \frac{\Lambda c^2}{3H_0^2} = \phi(R)^2 \Omega_{\Lambda_{\text{eff}},0},$$

where $\Omega_{\Lambda_{\text{eff}},0}$ denotes the cosmological component obtained by normalizing the geometric effective density $1/(R_c^2 \kappa)$ (corresponding to the cosmological-scale base strength $1/R_c^2$ multiplying the base response function $F(x)$ in $\Lambda_{\text{eff}}(r)$ as rewritten in Section 7) by the present critical density $\rho_{\text{crit},0}$, namely $\Omega_{\Lambda_{\text{eff}},0} \equiv (1/R_c^2) c^2 / (3H_0^2) = \Lambda c^2 / (3H_0^2)$. Therefore, the present value of the geometric resistive component is given as a cosmological component weighted by the geometric-response strength $\phi(R)^2$. Furthermore, since $\phi(R)^2 = 1/3$ at the characteristic scale $R = \sqrt{3}$ determined by the structure of the variational residual, we obtain

$$\Omega_{\text{geo},0}^{\text{resist}} = \frac{1}{3} \Omega_{\Lambda_{\text{eff}},0}.$$

That is, the present value of the geometric resistive component has a magnitude corresponding to one third of the cosmological component associated with Λ_{eff} . Moreover, as follows from Sections 4 and 7, Λ_{eff} is treated not as vacuum energy but as a geometric term, and therefore no independent fluid pressure has been postulated. Accordingly, at the background level, this component can be treated, in parallel with Ω_{b0} , as a pressureless matter-like effective component ($w = 0$), and its redshift dependence in the geometric effective density is given, in the standard manner, by

$$\rho_{\text{geo}}^{\text{resist}}(z) = \rho_{\text{geo},0}^{\text{resist}} (1+z)^3.$$

Here, using $x(z) = a(z) = (1+z)^{-1}$, this can be rewritten as

$$\rho_{\text{geo}}^{\text{resist}}(z) = \rho_{\text{geo},0}^{\text{resist}} x(z)^{-3},$$

and furthermore, since $x(z) = \ell_{\text{phys}}(z)/R$,

$$x(z)^{-3} = \left(\frac{R}{\ell_{\text{phys}}(z)} \right)^3.$$

It thus follows that the matter-like redshift dependence of the geometric resistive component is described by this geometric scale ratio. That is, the matter-like geometric resistive component that opposes the rise of the variational residual is characterized at the present epoch by $\phi(R)^2$, in the same manner as $\Lambda_{\text{eff}}(R)$, and in the background universe it rapidly decays, as $\ell_{\text{phys}}(z)$ increases with cosmic expansion, according to $x(z)^{-3}$, namely $\left(\frac{R}{\ell_{\text{phys}}(z)} \right)^3$. Then, the total matter component can be described as the sum of the baryonic matter component and the geometric matter-like component. Therefore, under the approximate flatness condition

$$\Omega_{r0} + \Omega_{b0} + \Omega_{\text{geo},0}^{\text{resist}} + \Omega_{\Lambda_{\text{eff}},0} = 1,$$

we obtain

$$\Omega_{\Lambda_{\text{eff}},0} = \frac{1 - \Omega_{r0} - \Omega_{b0}}{1 + \frac{1}{3}}.$$

Here, since the total matter component is

$$\Omega_{m0} = \Omega_{b0} + \Omega_{\text{geo},0}^{\text{resist}},$$

and since $\Omega_{\text{geo},0}^{\text{resist}} = \frac{1}{3} \Omega_{\Lambda_{\text{eff}},0}$, it follows that

$$\Omega_{m0} = \Omega_{b0} + \frac{1}{3} \cdot \frac{1 - \Omega_{r0} - \Omega_{b0}}{1 + \frac{1}{3}} = \Omega_{b0} + \frac{1 - \Omega_{r0} - \Omega_{b0}}{4}.$$

Therefore, substituting the representative values

$$\Omega_{b0} = 0.049, \quad \Omega_{r0} = 9 \times 10^{-5}$$

into the above expressions, we find quantitatively

$$\boxed{\Omega_{\Lambda_{\text{eff}},0} \simeq 0.71318, \quad \Omega_{\text{geo},0}^{\text{resist}} \simeq 0.23773, \quad \Omega_{m0} \simeq 0.28673}.$$

Using these values and following the same procedure as in Appendix A, we obtain

$$\boxed{z_c \simeq 0.08951}$$

as well as

$$\boxed{H_0^{\text{app}} \simeq 73.3 \text{ km s}^{-1} \text{ Mpc}^{-1}, \quad S_8 \simeq 0.758, \quad \Delta D_C(z_* = 1089.9) \approx +1.91 \times 10^{-6} \%}$$

From the above, it is shown that, under the approximate flatness condition, the geometric matter-like resistive component yields realistic cosmological parameters, and that $\Omega_{\text{geo},0}^{\text{resist}}$ can be interpreted as a matter-like restoring response that suppresses deviations from the variational extremum condition $\delta = 0$. In other words, this restoring response contributes to suppressing the rise of the finite-width, hill-like structure of the variational residual $\delta(r)$, while its present value remains only one third of $\Omega_{\Lambda_{\text{eff}},0}$, so that it cannot become dominant at the boundary $r = R$, where the transition from restoring resistance to geometric repulsive response begins; moreover, once r exceeds R , this resistance rapidly weakens. As a result, the geometric repulsive response is effectively released, and after passing over the hill, the system proceeds toward the region $r \gg R$, where $\delta(r) \rightarrow 0$ asymptotically.